

From classical to quantum optics: optics in a nutshell

Week

I

1 From Classical to Quantum Optics: Fundamentals

Looking at the landscape of physics and natural phenomena, we can say that light is, at the present moment, a rather well-understood subject. Indeed, the Maxwell's equations established a solid mathematical framework more than a century and describe - in a classical picture - the propagation of electromagnetic waves with remarkable success. With them, and with a suitable description of the material medium, we can account for most of the optical effects we encounter in our lives, including reflection, refraction, interference, diffraction, absorption, and many other.

At first sight, this makes the subject of quantum optics look slightly puzzling, and you may start asking yourself why you are here to learn quantum optics if Maxwell's equations already work so well.

Tackling this question in a conceptual manner, the answer is that the propagation of the electromagnetic field is only part of the problem. In many optical experiments, especially close to atomic or molecular resonances, the interesting physics lies in the response of matter to the field. That response can change very strongly with frequency and may involve effects that cannot be described adequately with a purely classical model of the material system. This introduces an approach we will call **semiclassical** - classical electromagnetic field, quantum model for matter - and that will be one of the major focuses of this course.

But we can also throw a second look: the electromagnetic field can also be quantized. Of course, it adds a layer of complexity that, for most optical problems is unnecessary, as a classical field interacting with quantum matter is already an excellent description. But there are a few situations where the discrete and statistical character of light itself may become essential, including analysis of noise, quantum interference effects, just to name a few. This quantum picture for light states is particularly important for building quantum technologies, and will also constitute a major focus of this course.

Therefore, putting into perspective, the aim of this curricular unit course

will not be to make you replace classical optics by quantum optics, but to make you build a new layer of knowledge through links between both classical and quantum pictures, fueled by the need to solve a given real-world problem.

For this objective, this first week begins with Maxwell's equations and the electromagnetic wave equation. We will look at some of their basic solutions and recover from them the usual language of ray and wave optics. Only then will we ask where this description starts to fail, and what additional ideas are required to describe light-matter interaction at the quantum level.

2 A brief historical overview

The nature of light - whether it behaves as a particle or a wave - is one of the most fascinating chapters of physics and a long discussion that spans centuries. This debate involves significant evolutions of scientific reasoning, a lot of new experimental techniques and evidence, and a very interesting history-politics-science crossover that even involves Napoleonic campaigns¹. Looking in retrospective, it can be said that there have been three big debates in the past about this topic, the first dating back to the 17th century and started by Isaac Newton and Christiaan Huygens.

Motivated by his successful interpretation of mechanics and forces, Newton advocated in his famous work *Opticks* (1703) that light was made up of corpuscles (small particles), explaining reflection and refraction with concepts closely related to classical mechanics. Yet, a decade before the publication of *Opticks*, Huygens² published *Traité de la Lumière* (1690), advocating that light was a wave, propagating layer by layer in the *ether* with each point in the wavefront acting as a spherical source - *Huygens Principle*. Now we know that Newton's ideas such as optical dispersion being related to particles of each color having distinct mass are far from being correct, while the Huygens Principle is still an interesting analogy to describe light propagation. But at that time Newton's immense influence in the academic world crystallized the vision of light for almost a century. The debate restarted in the 19th century with the works of Thomas Young and Fresnel. The double slit experiment by Young lectured in 1802 at the famous Royal Institution resulted in interference patterns that would only be explained if light behaved as waves. Although the double-slit sets a hallmark, Young was a polymath and a physician and never pursued a detailed mathematical formalism. This mathematical formalism was later achieved in France with Fresnel, who by the decade of 1810 was dedicated to refining the mathematical theory of Huygens, exploring light as a wave to explain more complex phenomena such as diffraction patterns around obstacles. Supported by the influential Arago, Fresnel established multiple mathematical methods based on wave behav-



Figure 1. Huygens and Newton, same hairstyle but distinct sides on the wave-particle debate.

¹ The book *"The History of Optical Interferometry and the Scientists Who Tamed Light"* by David Nolte presents a very interesting historical and scientific vision on the topic.

² Who also invented the pendulum clock of our grandfather's house.



Figure 2. Thomas Young, the polymath who shed the first light on deciphering wave optics and also the Rosetta stone.

ior which consolidated the wave theory for light. A few years later, and on the other side of the English Channel, the self-taught Michael Faraday was producing remarkable experimental results in the interplay of electric and magnetic fields. In particular, his discovery of the law of induction and later of the Faraday effect³ solidified a clear connection between both fields. These works inspired the work of the well-educated Maxwell, unifying electric and magnetic effects into a vector field formulation in his famous set of four partial differential equations. Besides, predicting the existence of electromagnetic waves that would propagate even in a vacuum, the classical theory for light that to the day we still use was essentially set. A third debate ignited later in the 20th century, with Planck, Einstein, and the advent of quantum mechanics. But we will come to that later wrapping up this first week.

2.1 Maxwell's Equations

The starting point of the classical description of light as an electromagnetic field are the Maxwell's equations, which relate the electric and magnetic fields to charges, currents, and to the response of the material through which the field propagates:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \quad (2.1)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (2.2)$$

$$\nabla \cdot \mathbf{D} = \rho, \quad (2.3)$$

$$\nabla \cdot \mathbf{B} = 0. \quad (2.4)$$

This set of equations explains the propagation of electromagnetic waves in terms of fields $\mathbf{E}$, $\mathbf{D}$, $\mathbf{B}$, and $\mathbf{H}$. While fields $\mathbf{E}$ and $\mathbf{H}$ correspond to electric and magnetic fields, the assisting fields $\mathbf{D}$ (electric displacement) and $\mathbf{B}$ (magnetic field, also called flux) contain information about how the material responds to the applied electric and magnetic fields. In specific, we can go to the microscopic properties of the material defining

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad (2.5)$$

and

$$\mathbf{B} = \mu_0 \mathbf{H} + \mathbf{M}, \quad (2.6)$$

where $\mathbf{P}$ is the electric polarization of the medium and $\mathbf{M}$ its magnetization.

During most of the curricular unit, we will simplify our approach with two strong yet solid assumptions:

- At optical frequencies, the electric interaction is normally much stronger

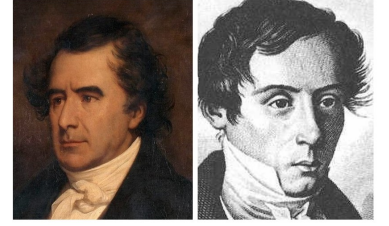


Figure 3. Arago and Fresnel, the French side.

³ in short, light polarization may be affected by a magnetic field.

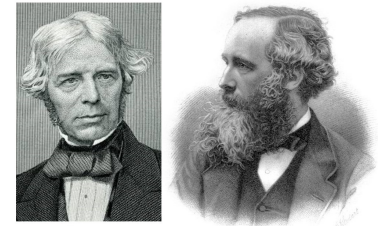


Figure 4. The experimentalist layman Faraday, and the theoretician Maxwell.

than the magnetic one, and for non-magnetic materials we will generally take

$$\mathbf{M} \simeq 0. \quad (2.7)$$

- For transparent insulating media away from electrodes or other sources we may usually neglect free currents and charges,

$$\mathbf{J} = 0, \quad \rho = 0. \quad (2.8)$$

These two assumptions not only simplify the framework, but already tells us something important about the problem we are trying to solve. Maxwell's equations determine how the electromagnetic field propagates once the response of the material is known. They do not, by themselves, tell us what that response is - this is the role of the $\mathbf{P}$ and $\mathbf{M}$ quantities. Therefore, during this course and within the above assumptions and the semiclassical approach, the important quantity that we will have to model will be material polarization $\mathbf{P}$.

We will get back to this next week, but for the moment, we will assume a simple linear model for the dielectric response, for which we may write

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \quad (2.9)$$

or equivalently

$$\mathbf{D} = \epsilon_0(1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}. \quad (2.10)$$

2.2 Wave equation, Waves, and Wavefronts

The two curl Maxwell equations allow us to explain in the macroscopic regime and far from the source how light can propagate in space as a field (the far field): a time-varying electric field leads to a time-varying magnetic field, which in turn produces a time-varying electric field again and again. Assuming that:

- ϵ and μ are spatially uniform and time constant (generally not true);
- There are no free charges $\rho = 0$ nor currents $\mathbf{J} = 0$;

we may take the rotational of the Faraday law to get

$$\nabla \times (\nabla \times \mathbf{E}) = -\nabla \times \frac{\partial \mathbf{B}}{\partial t} = -\frac{\partial}{\partial t} \nabla \times \mathbf{B} = -\epsilon_0 \mu_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} - \mu_0 \frac{\partial^2 \mathbf{P}}{\partial t^2}. \quad (2.11)$$

Then, by utilizing the vector calculus identity for the double rotational

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \quad (2.12)$$

and recovering the free charge assumption, i.e. $\nabla \cdot \mathbf{E} = 0$ and no relevant longitudinal variations of the material polarization (i.e. $\nabla \cdot \mathbf{P} = 0$) we end up with the wave equation for the electric field:

Electromagnetic wave equation:

$$\boxed{\nabla^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial^2 \mathbf{P}}{\partial t^2}} \quad (2.13)$$

where

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad (2.14)$$

is the speed of light in vacuum.

Equation 2.13 is particularly important because it separates the problem into two parts. The left-hand side describes the propagation of the electromagnetic field, while the polarization of the material appears on the right-hand side as a **source** for that field. Thus, once $\mathbf{P}$ is known, Maxwell's equations determine how the resulting optical field propagates in matter.

For the simple linear dielectric considered previously, $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$, and assuming $\epsilon = \epsilon_0(1 + \chi_e)$ is constant in space and time, one can work both equation 2.13 a bit further. Also, through a similar procedure, taking the curl of Ampère–Maxwell's law and using Faraday's law, gives the corresponding equation for the magnetic field. Therefore, for a linear, homogeneous, isotropic, non-magnetic and non-conducting medium we obtain:

Electromagnetic wave equations for linear media:

$$\boxed{\nabla^2 \mathbf{E} - \mu_0 \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0} \quad (2.15)$$

$$\boxed{\nabla^2 \mathbf{B} - \mu_0 \epsilon \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0.} \quad (2.16)$$

Both fields therefore propagate with the same wave velocity - light velocity -

$$v = \frac{1}{\sqrt{\mu_0 \epsilon}}, \quad (2.17)$$

which defines the refractive index

$$n = \frac{c}{v} = \sqrt{\frac{\epsilon}{\epsilon_0}} = \sqrt{1 + \chi_e}, \quad (2.18)$$

where $c = 1/\sqrt{\mu_0 \epsilon_0}$ is the light velocity in vacuum.

There are multiple possible solutions for the wave equation, being the most common in the context of optics, the plane wave solution, the spherical wave solution (far-field solution of a point-source, e.g. a pin-hole), and the cylindrical wave solution (far-field solution of a line-source, e.g. a slit). But before advancing with that discussion, let's call attention to three remarks that follow directly from the wave equation:

Light is a transverse wave in free space: This is easily obtained from $\nabla \cdot \mathbf{E} = 0$ and means that electric and magnetic fields propagate as transverse waves, i.e. the amplitude of both fields oscillate perpendicular to the direction of propagation defined by $\mathbf{k}$. This transverse nature normally holds in optical media but is not always true in special cases such as conductive media (related to the presence of plasmons), in the near-field (close to sources), and as guided or evanescent waves.

$\mathbf{E}$ and $\mathbf{B}$ are not independent: Indeed from the plane wave solution and taking the Faraday's law it is easy to demonstrate that

$$\mathbf{k} \times \mathbf{E} = \omega \mathbf{B} \quad (2.19)$$

meaning the two fields are perpendicular and $|\mathbf{E}| = \frac{c}{n} |\mathbf{B}|$. Note however this is only valid under the assumptions previously defined.

Superposition Principle: Assuming a linear response, the linearity of Maxwell's equations means that if $\mathbf{E}_1(\mathbf{r}, t)$ and $\mathbf{E}_2(\mathbf{r}, t)$ are solutions of the wave equation, then any linear combination

$$\mathbf{E}(\mathbf{r}, t) = a \mathbf{E}_1(\mathbf{r}, t) + b \mathbf{E}_2(\mathbf{r}, t) \quad (2.20)$$

is also a solution.

Definition. Energy Density and Poynting Vector: The energy density associated with an electromagnetic wave is given by

$$U = \frac{\mathbf{D} \cdot \mathbf{E} + \mathbf{B} \cdot \mathbf{H}}{2} \text{ (joules/m}^3\text{)}, \quad (2.21)$$

and the instantaneous flow of the field energy is given by the **Poynting Vector**

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \text{ (W/m}^2\text{)} \quad (2.22)$$

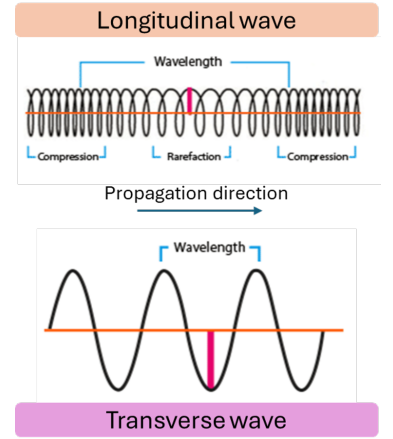


Figure 5. Light is a transverse wave, contrasting with longitudinal waves (e.g. sound).

From this we can also define the mean intensity as the

$$\langle S \rangle = I \text{ (W/m}^2\text{)} \quad (2.23)$$

2.2.1 Plane wave solution

To first introduce some basic definitions in the context of waves, let us first introduce the **plane wave solution**, a type of monochromatic harmonic wave solution that satisfies the wave equation ?? and is given by the ansatz⁴

$$\mathbf{E} = \text{Re}\{\mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t + \phi_0)}\}. \quad (2.24)$$

Using this ansatz we may introduce some formal definitions, namely:

- **Angular frequency of the wave** - $\omega = 2\pi f$
- **Wave frequency** - f ⁵
- **Wavenumber** - $k = 2\pi/\lambda$
- **Wavelength** - λ

To understand the language of optical waves, another crucial concept is the **Phase**. For a monochromatic plane wave with constant amplitude $\mathbf{E}_0$ ⁶, the phase is a scalar defined both in space and time that corresponds directly to the argument of the electric field phasor⁷, i.e.

$$\phi(\mathbf{r}, t) = \arg(\mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t + \phi_0)}) = \mathbf{k} \cdot \mathbf{r} - \omega t + \phi_0. \quad (2.25)$$

This allows to introduce two interesting concepts:

Wavefront - The wavefront corresponds to a set of points in space with same phase

$$\phi(\mathbf{r}, t) = \text{constant}, \quad (2.26)$$

oscillating in unison and corresponding to the same phase of oscillation. For example, for the simple case of a plane wave, it corresponds to a plane perpendicular to $\mathbf{k}$. Any deviations from the desired wavefront are called **aberrations**.

Phase Velocity - As the wavefront has temporal dependence, it is easy to conclude that constant phase surfaces moves in the propagation direction (in this case z) with a speed

$$v_p = \frac{\omega}{k}. \quad (2.27)$$

⁴ we opted for the $-\omega$ convention

⁵ f from 400THz to 700THz and λ from 700nm to 400nm for visible spectrum

⁶ typically the case you will encounter in this course

⁷ Phasor - representation of the electric field in the complex plane

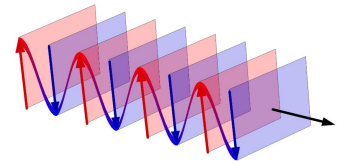


Figure 6. Plane wave has planar wavefronts (surfaces of constant phase, for a given time).

that we call phase velocity v_p . Note however, that this only tells you how fast crests move — not how fast energy moves.

Finally, the direct substitution of the ansatz leads to a relation between the wavelength or wavenumber of a wave to its frequency, which is called

Definition. Dispersion relation:

$$\omega = \frac{c}{n}k \quad (2.28)$$

2.3 Cylindrical Waves

Besides plane waves, another common wave solution arises when the source is extended along one axis (for example, an idealized line source, from a slit) and the natural symmetry becomes cylindrical. In cylindrical coordinates (ρ, ϕ, z) , a monochromatic outgoing solution of the scalar wave equation can be written as

$$\mathbf{E}(\rho, t) = \text{Re} \left\{ \mathbf{E}_0 \frac{e^{i(k\rho - \omega t)}}{\sqrt{\rho}} \right\}. \quad (2.29)$$

Wavefronts are thus cylindrical surfaces of constant ρ . Besides, the amplitude decays as $1/\sqrt{\rho}$, so the intensity decays as

$$I(\rho) \propto |\mathbf{E}|^2 \propto \frac{1}{\rho}, \quad (2.30)$$

which reflects the spread of energy in two transverse dimensions.

2.4 Spherical Waves

Another interesting wave solution family is the spherical wave, which is typically associated with the result of a point source in a homogeneous medium⁸. In spherical coordinates (r, θ, ϕ) , an outgoing monochromatic solution takes the form

$$u(r, t) = \text{Re} \left\{ A \frac{e^{i(kr - \omega t)}}{r} \right\}, \quad (2.31)$$

and wavefronts are thus spherical surfaces of constant r . The amplitude decays as $1/r$, and therefore the intensity decays as

$$I(r) \propto \frac{1}{r^2}. \quad (2.32)$$

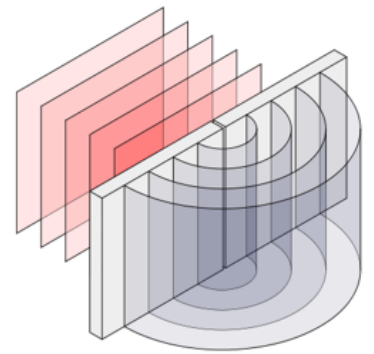


Figure 7. Cylindrical wavefronts.

⁸ e.g. an oscillating charge

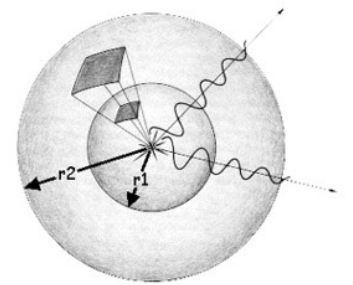


Figure 8. Spherical wavefronts.

2.5 Polarization

Up to this point, we have described wave propagation primarily through scalar solutions of the wave equation, focusing on phase, wavelength, dispersion, and spatial structure. However, electromagnetic waves are inherently vectorial: the electric and magnetic fields possess both magnitude and direction. The geometry of propagation is determined by the wavevector $\mathbf{k}$, but the field orientation within the plane transverse to $\mathbf{k}$ remains an additional and physically significant degree of freedom. This degree of freedom is known as *polarization*.

Taking the most common choice in optics $\mathbf{k} \parallel \mathbf{e}_z$ we may have:

- **Linear polarization:** if the field always oscillates along a given direction, e.g. $\mathbf{E} \parallel \mathbf{e}_x$;
- **Circular polarization:** if the field rotates once per cycle. A circular basis can be constructed as $\mathbf{e}_{R,L} = (1/\sqrt{2})(\mathbf{e}_x \pm i\mathbf{e}_y)$ with $\pm$ associated to right and left polarization states, respectively;
- **Elliptical polarization:** an extension of the circular polarization, but with different amplitudes for each cartesian component or phase difference $\theta \neq \pi/2$. A possible basis for this case is defined as $\mathbf{e}_{el,\pm} = (1/\sqrt{a^2 + b^2})(a\mathbf{e}_x \pm be^{i\theta}\mathbf{e}_y)$ with a and b being two amplitudes and θ a phase difference.

Overall, polarization plays no role in purely scalar diffraction theory, but it becomes essential when considering reflection and transmission (Fresnel coefficients), birefringent media, and in very special cases in light-matter interaction as we shall later explore.

3 Spatial Modes

While the monochromatic plane wave solution

$$\mathbf{E}(\mathbf{r}, t) = \text{Re} \left\{ \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right\}. \quad (3.1)$$

is mathematically convenient and locally accurate, it is also physically unrealistic. Considering $\mathbf{E}_0 = \text{constant}$ the ideal plane wave extends infinitely in space, and thus would correspond to an infinite total energy. But real optical beams have finite apertures and finite power and thus **must be described by fields with nontrivial transverse envelopes**. This naturally leads to the concept of **spatial modes**: structured field distributions that propagate in a stable manner and arise as eigensolutions of Maxwell equations under given boundary conditions.

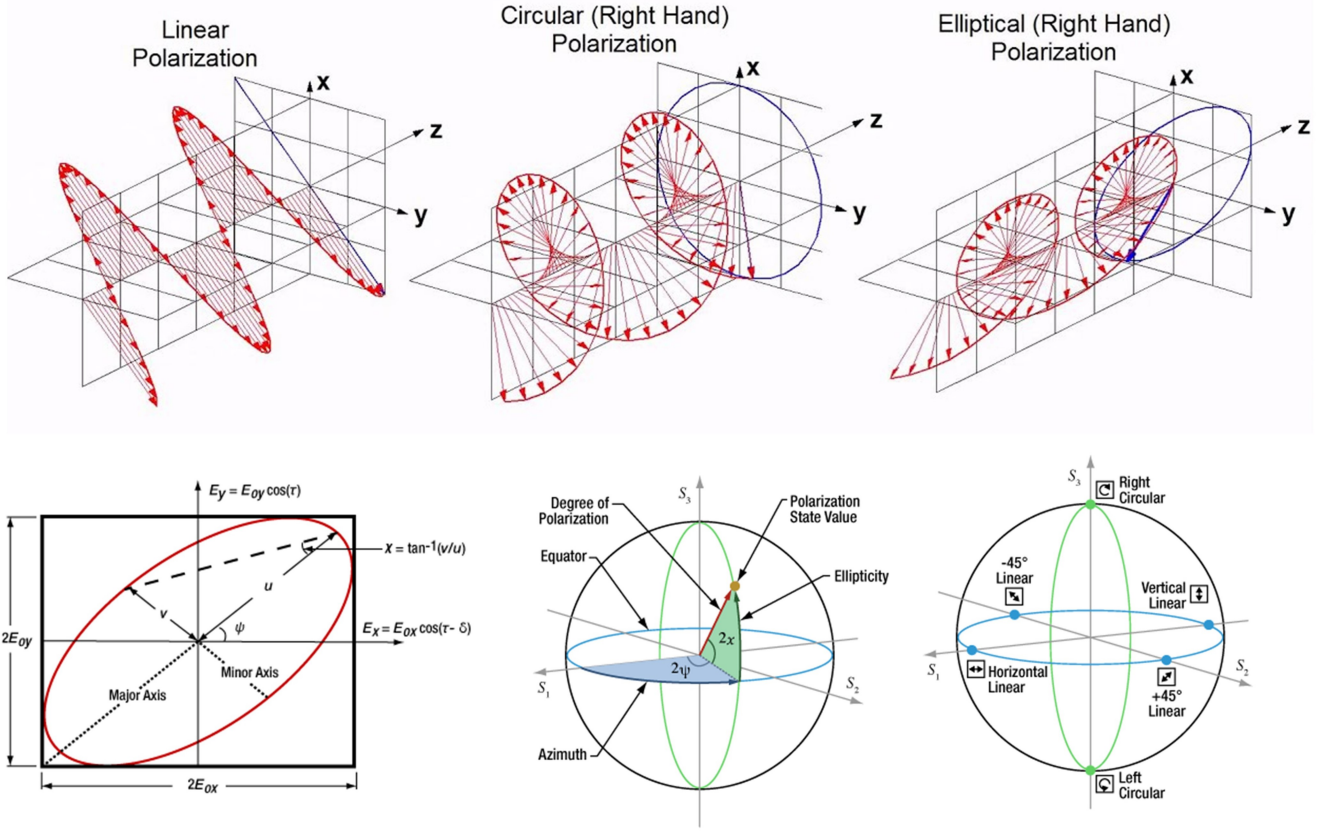


Figure 9. Visual representation of the distinct polarization states, demonstrating the absence or presence of rotation of the electric field transverse to the propagation direction. Note that in the complex field notation, this rotation along the z axis appears when you take the real value of the electric field. In the bottom row, we introduce a tool called **Poincaré Sphere** which allows us to easily identify the state of polarization(SOP) as a single vector, and possibly see some trajectories. It is a tool widely utilized in SOP sensing strategies.

To understand its origin, we start by assuming a monochromatic field (time-harmonic convention),

$$\mathbf{E}(\mathbf{r}, t) = \text{Re} \left\{ \mathbf{E}(\mathbf{r}) e^{-i\omega t} \right\} \quad (3.2)$$

in a linear, isotropic, non-magnetic dielectric ($\mu = \mu_0$) with refractive index $n(\mathbf{r})$. In source-free regions, Maxwell's equations reduce to a vector wave equation; a compact form for the electric field is

$$\nabla \times \nabla \times \mathbf{E}(\mathbf{r}) - k_0^2 n^2(\mathbf{r}) \mathbf{E}(\mathbf{r}) = 0, \quad k_0 = \frac{\omega}{c} = \frac{2\pi}{\lambda_0}. \quad (3.3)$$

In homogeneous media and under the usual transversality condition ($\nabla \cdot \mathbf{E} =$

0), this becomes the (vector) Helmholtz equation

$$\nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0, \quad k = nk_0. \quad (3.4)$$

To look at solutions of this equation we can try first to separate longitudinal propagation from transverse structure. For this, we **assume predominant propagation along** z and write (for a representative scalar component or a polarized field along $\mathbf{e}_p$)

$$E(\mathbf{r}) = u(x, y) e^{i\beta z}, \quad (3.5)$$

where $u(x, y)$ is the **transverse field distribution** and β is the **propagation constant**. Substituting Eq. (3.5) into a Helmholtz-type equation leads to an eigenvalue problem of the form

$$(\nabla_{\perp}^2 + k_0^2 n^2(x, y) - \beta^2) u(x, y) = 0, \quad (3.6)$$

with $\nabla_{\perp}^2 = \partial_x^2 + \partial_y^2$. The family of solutions $\{u_m(x, y)\}$ that satisfy the boundary conditions constitutes the **spatial modes**, each associated with a discrete β_m (guided systems) or a continuous spectrum (free space systems). In either case, a general field can be expanded as a superposition of modes, and describing the propagation largely reduces to the phase accumulation $e^{i\beta_m z}$ of each modal component.

3.1 Modes in free space and Gaussian beams

In many optical systems of interest (including most biomedical beam delivery and focusing), beams are well-collimated and propagate near the optical axis. This motivates the **paraxial approximation**, which is achieved by writing the field as a slowly varying envelope multiplying a carrier:

$$E(x, y, z) = u(x, y, z) e^{ikz}, \quad k = nk_0, \quad (3.7)$$

and assuming that u varies slowly with z compared to e^{ikz} . Substituting into the homogeneous Helmholtz equation and neglecting the $\partial_z^2 u$ term relative to $k\partial_z u$ yields the paraxial wave equation

$$\nabla_{\perp}^2 u + 2ik \frac{\partial u}{\partial z} = 0. \quad (3.8)$$

Equation (3.8) admits a family of structured solutions, with the most important being the fundamental Gaussian mode.

The fundamental Gaussian beam (also presented as TEM₀₀) can be writ-

ten as

$$u_{00}(r, z) = u_0 \frac{w_0}{w(z)} \exp\left(-\frac{r^2}{w^2(z)}\right) \exp\left[-i\left(\frac{kr^2}{2R(z)} - \psi(z)\right)\right], \quad (3.9)$$

with cylindrical symmetry $r^2 = x^2 + y^2$ and the standard beam parameters:

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2}, \quad (3.10)$$

$$R(z) = z \left[1 + \left(\frac{z_R}{z}\right)^2\right], \quad (3.11)$$

$$\psi(z) = \arctan\left(\frac{z}{z_R}\right), \quad (3.12)$$

$$z_R = \frac{\pi w_0^2}{\lambda}. \quad (3.13)$$

Here w_0 is the **beam waist** (corresponding to the minimum spot size) at $z = 0$, z_R is the **Rayleigh range**, $R(z)$ is the radius of curvature of the wavefront, and $\psi(z)$ is the Gouy phase, a hallmark of focused beams absent in plane-wave descriptions. Squaring to obtain the intensity, we get,

$$I_{gaussian}(r, z) = I_0 \left(\frac{w_0}{w(z)}\right)^2 \exp\left(-\frac{2r^2}{w^2(z)}\right) \quad (3.14)$$

The far-field divergence (half-angle, for $z \gg z_R$) is

$$\theta \simeq \frac{\lambda}{\pi w_0}, \quad (3.15)$$

expressing a fundamental trade-off: tighter focusing (small w_0) implies larger divergence. Note also that the Fourier transform of a Gaussian is still a Gaussian, meaning that the spatial shape is preserved both in the far field and in a focusing situation.

Putting into perspective, gaussian beams are much more than an analytical convenience. Indeed, they are the *natural output* of many stable laser resonators and the *natural acceptance* of single-mode fibers and diffraction-limited optical systems. In biomedical optics they appear repeatedly in context such as microscopy and micro-surgery (spot size and depth of focus are controlled by Gaussian beam parameters), confocal and multiphoton imaging, fiber delivery and endoscopy (coupling efficiency governed by spatial mode matching, speckle-reduction, further discussed in later sections).

Besides the fundamental gaussian mode, higher-order free-space modes (e.g. Hermite-Gaussian and Laguerre-Gaussian) also solve Eq. (3.8) and are relevant when shaping light (e.g., optical trapping, structured illumination),

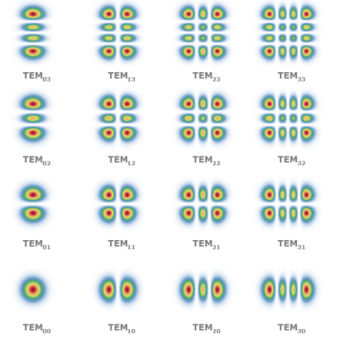


Figure 11. Format of Hermite Gaussian Modes.

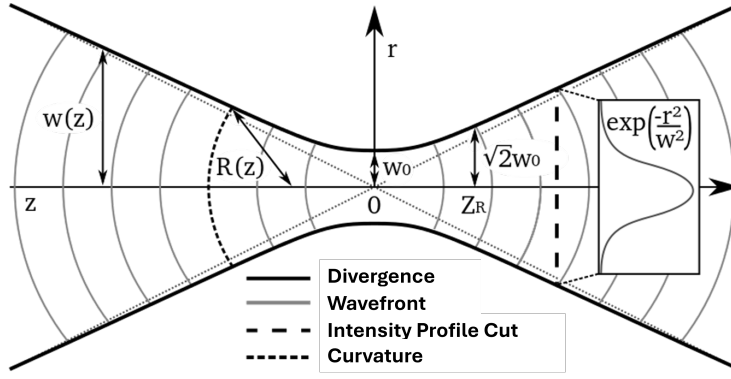


Figure 10. Gaussian mode and relevant parameters.

but the Gaussian fundamental mode can be considered the default in most of the applications you will cover in this course.

4 Ray optics and propagation

Having derived the electromagnetic wave equation (with representative solutions in homogeneous media), we shall now introduce *ray optics* as a controlled approximation of the full wave description. Ray optics is not a different theory of light⁹, but a *high-frequency* or *short-wavelength* limit of Maxwell's equations. **Thus, it is valid when the optical wavelength is much smaller than the characteristic length scales of the problem (e.g. curvature radii, aperture sizes, interface roughness, and spatial variations of refractive index).** In this regime, the phase varies rapidly while the field envelope varies slowly, and propagation can be described geometrically.

4.1 The eikonal approximation

As introduced before, the easiest ansatz is to consider a monochromatic (time-harmonic) field, $\mathbf{E}(\mathbf{r}, t) = \text{Re}\{\mathbf{E}(\mathbf{r})e^{-i\omega t}\}$, which reduces the wave equation to a Helmholtz-type equation in space

$$\nabla^2 \mathbf{E} - \frac{\omega^2 n^2}{c^2} \mathbf{E} = 0. \quad (4.1)$$

Ray optics then emerges directly from the Helmholtz equation by adopting the *eikonal ansatz* (WKB¹⁰ form),

$$\mathbf{E}(\mathbf{r}) = \mathbf{A}(\mathbf{r}) e^{ik_0 S(\mathbf{r})}, \quad (4.2)$$

where $k_0 = 2\pi/\lambda_0 = \omega/c$ is the vacuum wavenumber, $S(\mathbf{r})$ is a real phase function (the *eikonal*), and $\mathbf{A}(\mathbf{r})$ is a slowly varying vector amplitude. Sub-

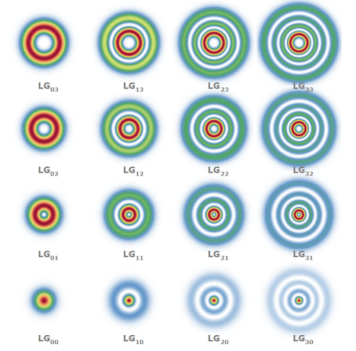


Figure 12. Format of Laguerre-Gaussian Modes.

⁹ *i.e. Light is still a wave*

¹⁰ *Wentzel-Kramers-Brillouin*

stituting into the Helmholtz equation and collecting terms by order in k_0 yields, at leading order k_0^2 , the *eikonal equation*

$$|\nabla S(\mathbf{r})|^2 = n^2(\mathbf{r}), \quad (4.3)$$

which links phase gradients to the refractive index distribution $n(\mathbf{r})$. The local wavevector is $\mathbf{k}(\mathbf{r}) = k_0 \nabla S(\mathbf{r})$, so rays are naturally defined as curves everywhere tangent to $\mathbf{k}$, i.e. normal to constant-phase surfaces (which we previously introduced as wavefronts). It is thus straightforward to conclude that in a uniform medium, Eq. (4.3) implies that ∇S is constant, and the rays are straight lines.

The function $S(\mathbf{r})$ has also a direct geometrical interpretation. Along a ray trajectory γ , its differential change satisfies

$$dS = n(\mathbf{r}) ds, \quad (4.4)$$

where ds is the differential arc length along the ray. Integrating along the path from point A to point B gives

$$S(B) - S(A) = \int_A^B n(\mathbf{r}) ds \equiv OPL, \quad (4.5)$$

defined as *optical path length* (OPL) between the two points and associated with a proportional accumulated phase of

$$\Delta\phi = k_0 OPL = \frac{2\pi}{\lambda_0} \int_A^B n(\mathbf{r}) ds. \quad (4.6)$$

4.2 Reflection and refraction at planar interfaces

Now that we know what to expect when light propagates in homogeneous media it is important to introduce what happens when light encounters a boundary between different optical media of distinct refractive index n_1 and n_2 . For that, we establish a coordinate system as presented in Figure 13, for which we choose the condition $k_i^y = 0$. In this coordinate system, we define the *plane of incidence* as the plane containing the incident wave vector $\mathbf{k}_i$ as well as the normal to the boundary.

When the ray encounters a change of index of refraction the direction changes, yet the phase must match continuously along the interface. In practice, this implies continuity of the *tangential* (i.e. $\parallel$, parallel to the interface) component of the wavevector,

$$(\mathbf{k}_i)_\parallel = (\mathbf{k}_r)_\parallel = (\mathbf{k}_t)_\parallel, \quad (4.7)$$

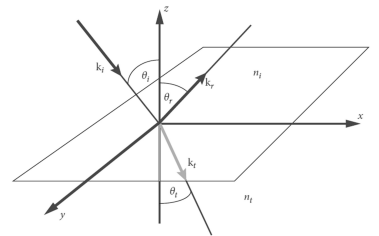


Figure 13. Coordinate system for computing the reflection and refraction angles.

This immediately implies that all three rays lie in the same plane (the plane of incidence), spanned by the incident ray and the surface normal.

Reflection. For a reflection, both the incident and reflected rays propagate in medium 1, so

$$|\mathbf{k}_i| = |\mathbf{k}_r| = k_0 n_1. \quad (4.8)$$

Equation (4.7) enforces identical tangential components, hence the normal component must reverse sign. Geometrically this yields

Reflection Law:

$$\theta_r = \theta_i. \quad (4.9)$$

Refraction. For transmission into medium 2,

$$|\mathbf{k}_i| = k_0 n_1, \quad |\mathbf{k}_t| = k_0 n_2. \quad (4.10)$$

Taking the magnitudes of the tangential components in Eq. (4.7) gives the following.

$$|\mathbf{k}_i| \sin \theta_i = |\mathbf{k}_t| \sin \theta_t, \quad (4.11)$$

and therefore we obtain

Snell's Law:

$$n_1 \sin \theta_i = n_2 \sin \theta_t, \quad (4.12)$$

where θ_i and θ_t are measured from the normal surface. Note that when $n_1 > n_2$, Eq. (4.12) implies that a propagating transmitted ray exists only if $\sin \theta_t \leq 1$, i.e. only if $\theta_i \leq \theta_c$, where

$$\sin \theta_c = \frac{n_2}{n_1}. \quad (4.13)$$

For $\theta_i > \theta_c$ there is no propagating transmitted ray, and thus we have total internal reflection, a key piece of guided-wave optics.¹¹

Amplitude and Phase. We now know what happens to the direction of the rays, however we still do not know what happens to phase and amplitude of the reflected and refracted wave. Relations for reflectivity and refraction are commonly known as Fresnel's laws, and for a medium free of charges and currents, they can be derived from the continuity of the tangential components of E and H , taking into account the polarization direction. For that, we will define a p-parallel polarization and s-perpendicular polarization according to the coordinate system of Figura 17. In these notes, we will merely state the Fresnel laws and consider their principal physical impact; exact derivations can be found elsewhere¹², and

¹¹ wave optics shows that the transmitted field becomes evanescent in medium 2 and the reflection is (ideally) lossless.

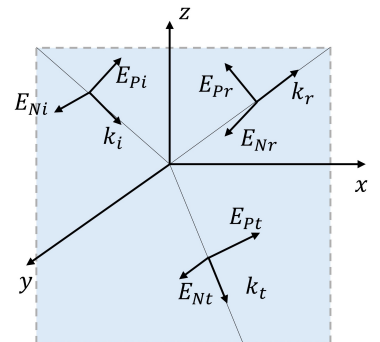


Figure 14. Coordinate system and plane of incidence.

¹² e.g. Hecht

Fresnel Laws

S (Perpendicular) polarization: For perpendicular polarization (to the plane of incidence) we only need to consider the two equations involving E_N terms. For transmission, it can be shown that

$$t_s = \frac{E_{Nt}}{E_{Ni}} = \frac{2}{1 + \frac{\mu_1 \tan \theta_i}{\mu_2 \tan \theta_t}} \approx \frac{2 \sin \theta_t \cos \theta_i}{\sin(\theta_i + \theta_t)} \quad (4.14)$$

with the approximation valid for $\mu_1 \approx \mu_2$. For the reflection coefficient we have

$$r_s = \frac{E_{Nr}}{E_{Ni}} = \frac{-\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)}. \quad (4.15)$$

In particular, note that a π phase shift is observed in the reflected wave if the index of refraction satisfies $n_2 > n_1$.

P (Parallel) polarization: For parallel polarization (to the plane of incidence) we only need to consider the two equations involving E_t terms. It can be shown that

$$t_p = \frac{E_{pt}}{E_{pi}} = \frac{2 \cos \theta_i \sin \theta_t}{\cos \theta_t \sin \theta_t + \frac{\mu_1}{\mu_2} \cos \theta_i \sin \theta_i} \quad (4.16)$$

for the transmission coefficient and

$$r_p = \frac{E_{Pr}}{E_{Pi}} = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} \quad (4.17)$$

Note that if $\tan(\theta_i + \theta_t) = +\infty$ we may obtain no reflection for the parallel polarization. This situation is called the *Brewster's angle*.

5 Wave Optics: Diffraction and Interference

Geometric optics describes light propagation in terms of rays that follow trajectories determined by Snell's law and Fermat's principle. As stated, this approximation is valid when the wavelength is much smaller than all relevant geometrical scales. However, ray optics cannot describe two important effects that we observe in real-world scenarios: diffraction and interference. For that, we need a wave description of the propagation.

5.1 Wave Propagation as Superposition

In wave optics, propagation is not described by individual trajectories but by the coherent superposition of secondary wavelets. According to the Huygens–Fresnel principle, every point of a wavefront acts as a source of spherical

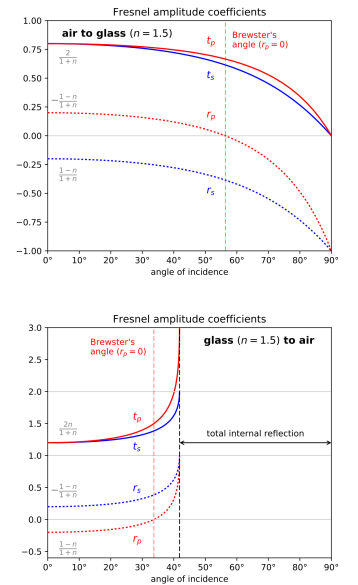


Figure 15. Fresnel amplitude coefficients.

waves, and the field at a later position is obtained by summing the contributions from all points on the previous wavefront.

Near-field regime. Considering the monochromatic ansatz wave solution $\mathbf{E} = u(\mathbf{r})e^{-i\omega t}\mathbf{e}_p$ polarized along $\mathbf{e}_p$, we have that the scalar field $u(\mathbf{r})$ shall satisfy the Helmholtz equation

$$\nabla^2 u(\mathbf{r}) + k^2 u(\mathbf{r}) = 0. \quad (5.1)$$

Under the **paraxial approximation** - i.e. assuming that forward propagation predominantly occurs along the z -axis - it is possible to show that the field at a plane located a distance z from an aperture plane can be approximately obtained from a Fresnel diffraction integral

$$u(x, y, z) = \frac{e^{ikz}}{i\lambda z} \iint u(x', y', 0) \exp\left[\frac{ik}{2z}((x - x')^2 + (y - y')^2)\right] dx' dy'. \quad (5.2)$$

This expression shows explicitly that propagation is an integral operation: the field at each point results from coherent contributions of all points in the aperture. Equation (5.2) describes propagation in the *near-field* or Fresnel regime. In this regime, quadratic phase terms in both the source coordinates (x', y') and observation coordinates (x, y) are retained. The diffraction pattern evolves continuously with propagation distance.

Far-field regime. When the observation distance z is large, the quadratic terms in (x', y') inside the exponential become negligible compared to the cross terms. The propagation integral simplifies to

$$u(x, y, z) \propto \iint u(x', y', 0) \exp\left[-i\frac{2\pi}{\lambda z}(xx' + yy')\right] dx' dy'. \quad (5.3)$$

and thus the pattern in the far-field is proportional to the two-dimensional Fourier transform of the aperture field with spatial frequencies given by

$$f_x = \frac{x}{\lambda z}, \quad f_y = \frac{y}{\lambda z}. \quad (5.4)$$

Under this condition, diffraction patterns directly encode the spatial-frequency content of the aperture. For example, a single slit of width a produces a far-field intensity pattern of the form

$$I(\theta) \propto \left(\frac{\sin \beta}{\beta}\right)^2, \quad \beta = \frac{\pi a}{\lambda} \sin \theta \approx \frac{\pi a x}{\lambda z}, \quad (5.5)$$

whose angular width scales as $\Delta\theta \sim \lambda/a$. A narrower slit therefore produces a broader diffraction pattern. This behavior is a direct manifestation

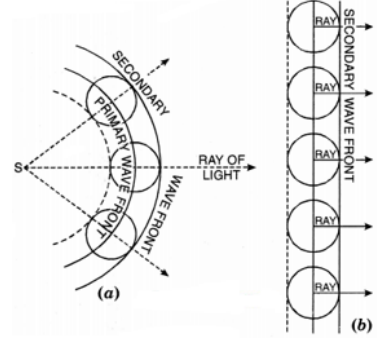


Figure 16. Huygens principle - each point of a wavefront acts as a source.

of the Fourier relationship between aperture size and angular spread. By contrast, ray optics predicts that light passing through a slit should propagate in straight lines and produce a sharp geometric shadow with width equal to the slit.

All in all, diffraction shall be understood as a fundamental consequence of the wave nature of light and not as a correction to geometric optics. It ultimately sets the limits of focusing, imaging, and spatial resolution in biomedical optical systems, as we will see later in the course.

5.2 Interference

The superposition principle warrants that the sum of two solutions of the wave equation is also a solution of the wave equation itself. While the effects of this overlap are not seen in the field directly, a square law detector¹³ reveals bright and dark bands called **fringes** which we could relate to the concept of **interference**.

Interference occurs when two or more coherent light waves (waves of the same frequency and constant phase difference) overlap. In particular, by definition, we say that *constructive interference* occurs where the waves are in phase (the crest of one wave meets the crest of another), leading to increased light intensity, whereas waves out of phase (the crest of one wave meets the trough of another) results in *destructive interference* with decreased intensity.

To understand the basics of wave interference we may consider the simple case of two plane waves in the complex notation,

$$\mathbf{E} = E_1 e^{i(-\omega_1 t + \mathbf{k}_1 \cdot \mathbf{r}_1 + \phi_1)} \mathbf{e}_1 + E_2 e^{i(-\omega_2 t + \mathbf{k}_2 \cdot \mathbf{r}_2 + \phi_2)} \mathbf{e}_2 \quad (5.6)$$

where $\mathbf{e}_i$ is a unit polarization vector for each field and $\mathbf{r}_i$ the spatial position where the amplitude is to be added in relation to the source. The intensity of the wave can be calculated as

$$I = \frac{cn\epsilon_0}{2} |\mathbf{E}|^2 \quad (5.7)$$

$$= \frac{cn\epsilon_0}{2} (\mathbf{E}_1 \cdot \mathbf{E}_1^* + \mathbf{E}_2 \cdot \mathbf{E}_2^* + 2 \operatorname{Re}\{\mathbf{E}_1 \cdot \mathbf{E}_2^*\}) \quad (5.8)$$

$$= I_1 + I_2 + 2 \frac{cn\epsilon_0}{2} \operatorname{Re}\{E_1 E_2^* e^{i((\omega_2 - \omega_1)t + (\mathbf{k}_1 \cdot \mathbf{r}_1 - \mathbf{k}_2 \cdot \mathbf{r}_2) + \phi_1 - \phi_2)} \mathbf{e}_1 \cdot \mathbf{e}_2^*\} \quad (5.9)$$

$$= I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\Delta\omega t + (\mathbf{k}_1 \cdot \mathbf{r}_1 - \mathbf{k}_2 \cdot \mathbf{r}_2) + \Delta\phi) \operatorname{Re}\{\mathbf{e}_1 \cdot \mathbf{e}_2^*\} \quad (5.10)$$

and the interference signal is contained in the last term of the above equation, where $\Delta\omega_{21} = \omega_2 - \omega_1$ and $\Delta\phi = \phi_2 - \phi_1$. Some considerations:

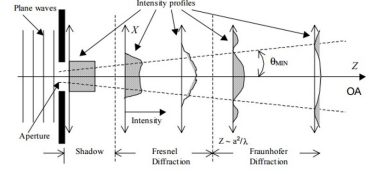


Figure 17. Far-field pattern generated for a small slit, comparing with the ray optics prediction.

¹³ a detector which outputs a signal proportional to the intensity.

- If the polarizations are collinear the interference is maximized, whereas if perpendicular there will be no interference.
- The dependency on time is trickier than it seems at first sight. First, in real-world conditions ϕ_2 and ϕ_1 may change with time (stochastic phase jumps), meaning that for obtaining a proper interferometric signal special care must be taken concerning the coherence length. Also, for distinct time frequencies, a beating may appear and its detection will strongly depend on the frequency shift $\Delta\omega_{21}$. In the most trivial case of the same frequencies, this is not an issue.

For the next sections, we will consider the case of interference of waves with same frequency and polarization and assume that over a given time period (integration time or exposure) the phase difference $\Delta\phi = \phi_1 - \phi_2$ maintains constant, for which we obtain the temporal average $\langle \cdot \rangle$ as

$$\langle I \rangle = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\mathbf{k}_1 \cdot \mathbf{r}_1 - \mathbf{k}_2 \cdot \mathbf{r}_2 + \Delta\phi) \quad (5.11)$$

So far we disregarded the fact that the envelope function $E_i(x, y, z)$ may also contain a transverse spatial distribution. If that is the case, an explicit dependence will appear and the intensity of the interference may vary due to the variation in the intensity of each wave, which may affect visibility. This can be even more problematic in the presence of multiple wave interference or many modes as it happens in multimode waveguides.

Definition. Contrast or visibility in interference: The contrast of interference fringes may be defined as a normalized value between 0 and 1 as

$$Contrast = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}. \quad (5.12)$$

The contrast depends on multiple factors, with the most important being the **polarization state**, the **coherence of the source**, the **amplitude** of the waves, and the **spatial modes or envelope functions**.

Finally, note that it is not problematic to add a spatial distribution to the phase difference, which ultimately establishes the principles of digital off-axis holography.

5.3 Young Double-Slit

Although Newton's Rings and thin film interference are both manifestations of interference and wave-like behavior of light, they were not envisioned nor realized to probe that behavior at the scientific level. In this context, the double-slit experiment, conducted for the first time by Thomas Young in

the early 19th century, was a **paradigm shift**. Indeed, the experiment was designed to demonstrate the wave nature of light through the phenomenon of interference.

In general terms, the experiment is based on the principle that light, when treated as a wave, exhibits constructive and destructive interference under certain conditions. Indeed, when coherent light waves from two closely spaced slits converge on a screen, they superimpose on each other, creating an interference pattern.

To model this, let us assume that each slit acts as a source, meaning that following our framework introduced previously, we have

$$\mathbf{E}(\mathbf{r}) = E_1(\mathbf{r})e^{i(-\omega t + \mathbf{k} \cdot \mathbf{r}_1 + \phi_1)}\mathbf{e}_p + E_2(\mathbf{r})e^{i(-\omega t + \mathbf{k} \cdot \mathbf{r}_2 + \phi_2)}\mathbf{e}_p \quad (5.13)$$

where $\mathbf{e}_p$ is the polarization vector and $\mathbf{r}_1$ and $\mathbf{r}_2$ the distances of the spatial position $\mathbf{r}$ to the source. Assuming the relative phase $\Delta\phi = \phi_2 - \phi_1 = 0$ we get that the interference detected at a plane located at $z = L$ (see figure 18 for coordinates) is given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(k(r_1 - r_2)). \quad (5.14)$$

For sufficiently small angle θ (i.e. $L \gg y$) one can consider the two vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ approximately parallel and the path difference becomes simply

$$r_1 - r_2 \approx d \sin\theta = \frac{dy}{L}. \quad (5.15)$$

Depending on the difference in path lengths traveled by the waves from each slit to a point on the screen, the interference can be either constructive (bright fringe) if

$$y = \frac{m\lambda L}{d} \quad (5.16)$$

for m integer or destructive (dark fringe) for

$$y = \frac{(m + 1/2)\lambda L}{d}. \quad (5.17)$$

Putting into broader perspective, the double-slit experiment underscores several key concepts in interferometry that we will utilize in the next few weeks:

- **Wavefront Division:** by dividing the wavefront and recombining it later, you can obtain interference (if within the spatial and temporal coherence parameters of your source). This property will be utilized in many different interferometers during the next few weeks;

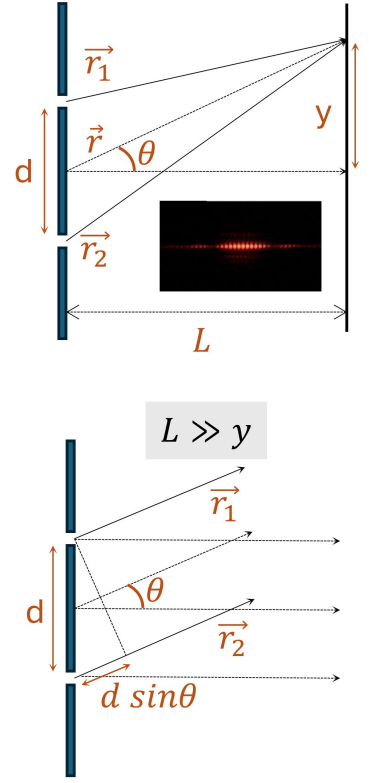


Figure 18. Young's double slit experiment, coordinate system and approximation.

- **Coherence of light:** contrary to Newton's ring and thin film interference, double slit wave interference is only possible when using spatially coherent light sources, meaning that light sources are not the same, nor lead to the same results;
- **Path difference:** The extreme sensitivity of the interference pattern to the path difference between the two waves illustrates how interferometers can be used to detect minuscule changes in distance or optical properties of materials.

As seen, the double-slit experiment sets the cornerstone of wave-like behavior of light and interference. Understanding the principles, framework, and details behind the double-slit experiment is essential for a young student engaged in the study of interferometry and ultimately physics. Indeed, this simple yet remarkable demonstration encloses one of the most paradigm-shift ideas of physics, which about two centuries later reshaped again the landscape of physics in the form of quantum mechanics.

6 Light as a Particle: Photons and Detection

Up to this point, light has been described as a classical electromagnetic wave. This framework successfully explains propagation, reflection, refraction, diffraction, interference, and polarization. However, when light is generated, absorbed, or detected at sufficiently low intensities, its discrete nature becomes essential. In these regimes, light behaves as a stream of quanta we call *photons*.

The need for a quantized description of radiation emerged at the beginning of the twentieth century. Classical electrodynamics failed to describe the spectrum of blackbody radiation, leading to the so-called ultraviolet catastrophe. In 1900, Planck resolved this paradox by proposing that energy exchange between matter and radiation occurs in discrete units proportional to frequency. Shortly thereafter, Einstein explained the photoelectric effect by postulating that light itself is composed of energy quanta, building on top of the quantized view of Planck. These developments established a crucial idea: **electromagnetic fields propagate according to Maxwell's equations, but energy is exchanged with matter in discrete packets.**

For this curricular unit, a particular relevant observation is that:

Photon Energy and Momentum: a photon of angular frequency ω carries energy

$$E = \hbar\omega = hf, \quad (6.1)$$

and momentum

$$p = \hbar k = \frac{h}{\lambda}. \quad (6.2)$$

These relations connect the wave parameters (ω, k) to particle-like quantities (E, p) , **with the frequency determining the energy delivered to matter.**

For biomedical optics, the key idea is that while light propagation through tissue is often accurately described by classical wave theory, absorption, fluorescence, and photochemical interactions occur through the exchange of individual photons. The amount of energy deposited in tissue depends directly on photon energy, and optical detection ultimately consists of discrete absorption events. Yet, note that both pictures are still complementary aspects of a unified physical theory, and biomedical optics routinely operates at the interface between these descriptions - wave theory is used to design and model optical systems, while the photon picture is utilized to understand energy transfer, excitation processes, and detection.